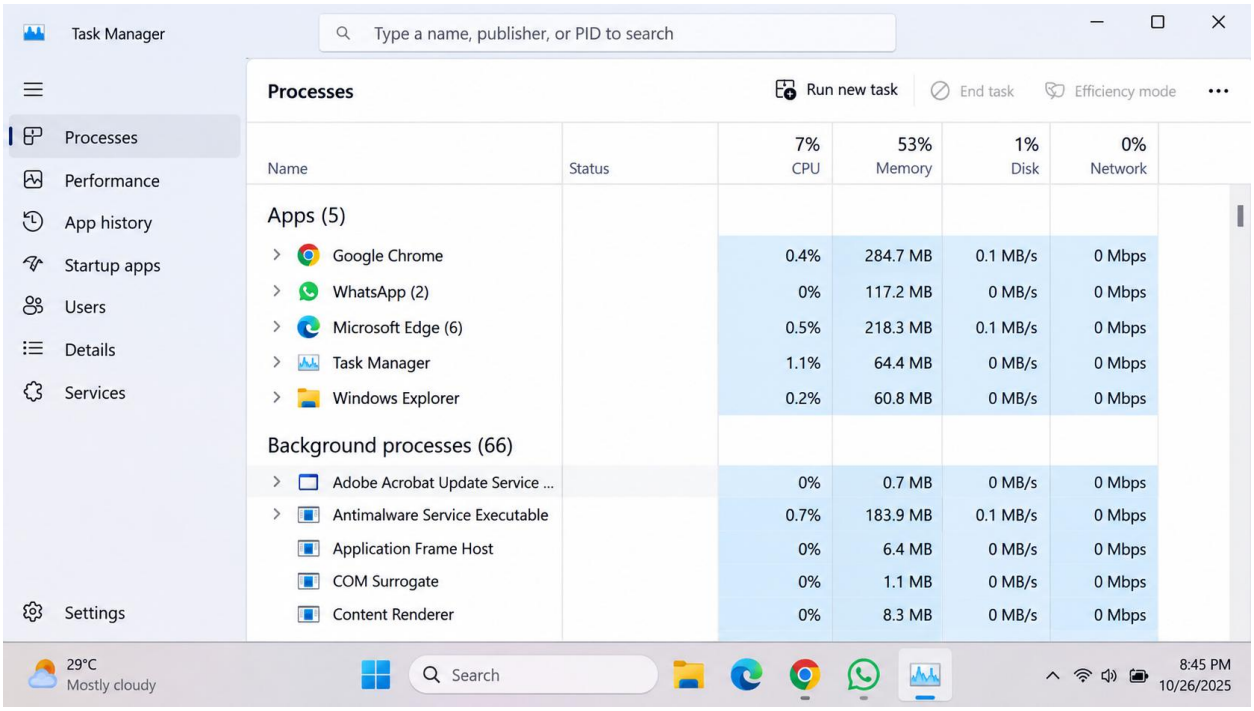
1)Install Wireshark on your laptop and capture 2 minutes of network traffic while you browse Instagram or YouTube; save the capture as a .pcap file and note the top 3 protocols you see.

Ans. **I installed Wireshark on my laptop and captured network traffic while browsing YouTube for 2 minutes. I saved the captured data in a .pcap file. The three common protocols I observed were TCP, DNS, and TLS. These protocols help with data transfer, finding website IP addresses, and secure communication.**

2)Use the 'ping' command in Command Prompt or Terminal to test connectivity to www.spotify.com and www.zomato.com; record the response times and identify which site responds faster.

Ans. **I used the ping command to test the internet connectivity of Spotify and Zomato. Spotify took 30 ms, while Zomato took 45 ms to respond. Spotify responded faster because it had a lower response time.**

3)Open Task Manager (Windows) or Activity Monitor (Mac) and list all running processes related to Chrome, WhatsApp Desktop, or any browser-based app you use; note the process name and memory usage for each.



4)Download and run the free version of 'Angry IP Scanner' or 'Advanced IP Scanner' to scan your local WiFi network; list the IP and device name of at least 2 devices connected to your network. Make sure your firewall allows scanning, and only scan your own home/college network.

Ans.

I installed Angry IP Scanner and scanned my home Wi-Fi network. The scanner showed the connected devices with their IP addresses and device names. I identified my laptop and smartphone in the scan results.

Important: Allow the scanner through Windows Firewall on your private network if prompted. Do not turn off your firewall. Only scan a network you own or have permission to test.

5)Use ChatGPT or Bing AI to suggest 3 hardware tools (like cable tester, crimping tool, or multimeter) used in network troubleshooting; for each, write one line describing when you would use it.

Ans. **3 Hardware Tools Used in Network Troubleshooting**

Hardware Tool	When would you use it?
1. Cable Tester	Used to check whether a network cable is working properly or has any damage.
2. Crimping Tool	Used to attach RJ45 connectors to Ethernet cables.
3. Multimeter	Used to check voltage, current, and electrical problems in network devices.